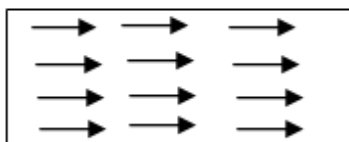




Q1. What are permanent magnets? Give one example.

Answer: A permanent magnet is made up of material which when magnetized retains magnetic field for longer period and has high value of coercivity and remanence or retentivity. The magnetic moment of atoms gets aligned permanently in such a way that resultant magnetic moments of all atoms gives persistent magnetic field even after removal of external magnetic field.

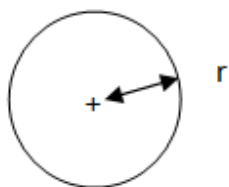


In a permanent magnet, higher magnetic flux obtained with smaller volume of material.

Example: Alloys of aluminium nickel and cobalt (Alnico)

Q2. What is the geometrical shape of equipotential surfaces due to a single isolated charge?

Answer: All points on a sphere around a single isolated charge have same distance r so all have same potential.



Thus each point on the surface of the sphere is equidistant from the charge, therefore geometrical shape of equipotential surface will be concentric sphere such that potential inversely proportional to distance (r).

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$



Q3. Which of the following waves can be polarized (i) Heat waves (ii) Sound waves?
Give reason to support your answer.

Answer: Heat waves being electromagnetic wave exhibits polarisation since electric vector oscillates perpendicular to the direction of propagation in all direction. Sound waves cannot be polarised as the direction of oscillation is along the direction of propagation.

Q4. A capacitor has been charged by a dc source. What are the magnitudes of conduction and displacement currents, when it is fully charged?

Answer: When the capacitor is charged initially charge flows and potential of capacitor increased, during charging we will get current called conduction current. Therefore during charging voltage is changing, electric field changes which changes electric flux in the dielectric of the capacitor hence electric flux current density that is displacement current obtained till it is not fully charged.

Once capacitor is fully charged, potential across the plates of capacitor becomes equal to the potential of dc source hence conduction and displacement current both becomes zero.

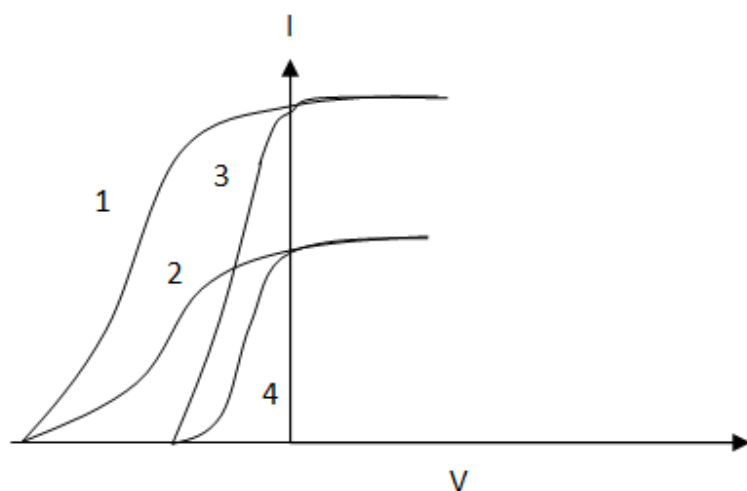
Q5. Write the relationship between angle of incidence 'i', prism 'A' and angle of minimum deviation for a triangular prism.

Answer: Angle of incidence (i), is related to angle of minimum deviation (δ_m) and angle of prism (A) as

$$i = \frac{A + \delta_m}{2}$$



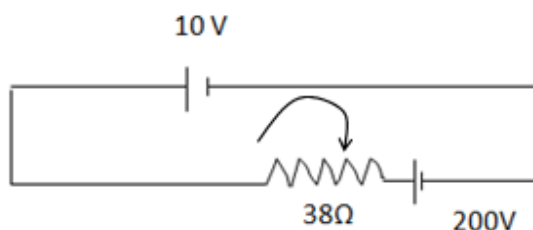
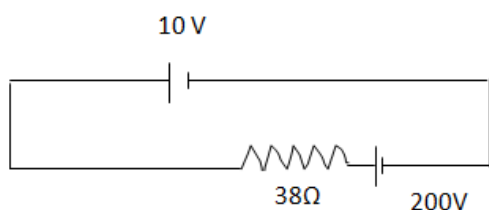
Q6. The given graph shows the variation of photo-electric current (I) versus applied voltage (V) for two different photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials but same intensity of incident radiation.



We know for different material stopping potential may be different. Therefore graph starting from same point on X-axis i.e. stopping potential is unlikely to be of different material.

We know that with increase in V , current I increased if intensity of incident light kept constant. Therefore from the graph obvious correct pairs (1,3) and (2,4) corresponds to different materials having same intensity of incident radiation.

Q7. A 10 V battery of negligible internal resistance is connected across a 200 V battery and a resistance of $38\ \Omega$ as shown in the figure. Find the value of the current in circuit.



Considering clock wise direction as positive, applying Kirchhoff's loop rule. Let I be the current in the circuit then

$$-10 + 200 - 38I = 0$$

$$\text{or } I = 190/38 = 5\ \text{A}$$



Q8. The emf of a cell is always greater than its terminal voltage. Why? Give reason.

Answer: emf is the potential difference of the cell terminal when no current is drawn from the cell. If current drawn from the cell say I and internal resistance of the cell be r then because of this internal resistance of the cell there will be a potential drop of Ir , hence terminal voltage will be reduced by Ir .
I.e. $V = E - Ir$ therefore $E > V$.

Q9. (a) Write the necessary conditions for the phenomenon of total internal reflections to occur.

Answer:

1. Light must travel from denser medium to rarer medium
2. Angle of incidence must be more than critical angle.

(b) Write the relation between the refractive index and critical angle for a given pair of optical media.

Answer:

	We know that product of Refractive index and sine of angle of incidence is constant. Therefore $\mu_d \sin \theta_c = \mu_r \sin 90^\circ \quad \text{or} \quad \sin \theta_c = \frac{\mu_r}{\mu_d}$
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Q10. State Lenz's Law. A metallic rod held horizontally along east-west direction, is allowed to fall under gravity, will there be an emf induced at its ends? Justify your answer.

Answer: Lenz's Law

Statement: "The direction of induced current or emf in a circuit is always such that it opposes the cause which produces it."

It gives the direction of current or emf induced in a circuit.

We know that earth's magnetic field along N-S direction, here velocity of the rod is along downward direction, magnetic field is cutting the plane of the metallic rod hence because of em induction emf will be induced.



Q11. A convex lens of focal length 25 cm is placed coaxially in contact with a concave lens of focal length 20 cm. Determine the power of the combination. Will the system be converging or diverging in nature?

Answer: We know that power of two lenses kept in contact is given by $P = P_1 + P_2$

Here $f_1 = +25 \text{ cm} = +25/100 \text{ m} = 1/4 \text{ m}$ hence $P_1 = 1/f_1 = +4 \text{ Dioptre}$,

$f_2 = -20 \text{ cm} = -20/100 = 1/5 \text{ m}$, hence $P_2 = 1/f_2$

Therefore power of the combination $P = +4 - 5 = -1 \text{ Dioptre}$

Since Power is negative hence system will behave as diverging in nature.

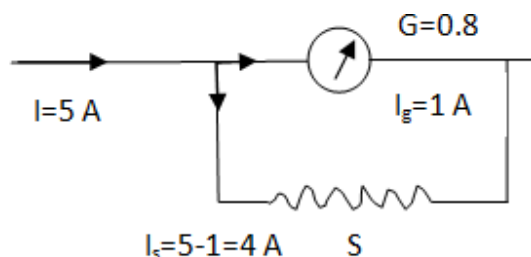
Q12. An ammeter of resistance 0.80Ω can measure current up to 1.0 A .

(i) What must be the value of shunt resistance to enable the ammeter to measure current up to 5.0 A ?

(ii) What is the combined resistance of the ammeter and the shunt?

Answer:

We know for shunt resistance is connected in parallel to ammeter. Here G =Ammeter Resistance, S =Shunt resistance, I_g =Current through Ammeter, I_s =Current through Shunt as shown



(i) To measure higher current shunt resistance(S) to be connected in parallel to Ammeter.

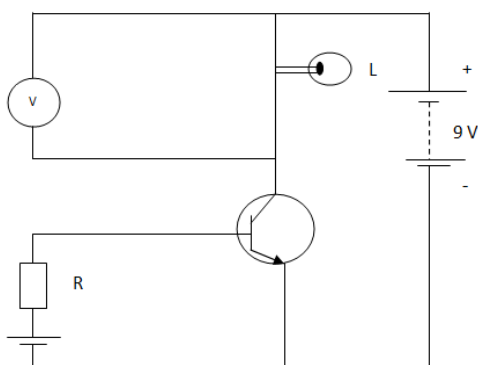
From the diagram, $S \times I_s = G \times I_g$ or $S \times 4 = 0.8 \times 1$ or $S = 0.8/4 = 0.2 \Omega$

(ii) Shunt resistance and Ammeter resistance in parallel there equivalent resistance =

$$R = \frac{R \times S}{R + S} = \frac{0.8 \times 0.2}{0.8 + 0.2} = 0.16 \Omega$$



Q13. In the given circuit diagram, a voltmeter 'V' is connected across a lamp 'L'. How would (i) the brightness of the lamp and (ii) voltmeter reading 'V' be affected, if the value of resistance 'R' is decreased? Justify your answer.



- (i) In the transistor circuit if R decreased, base current I_b will increase, but the gain of the transistor $\beta = \frac{I_c}{I_b}$, if I_b increases collector current I_c will also increase to keep β same. As more current I_c flows to lamp, power dissipated by lamp will increase hence brightness of the lamp will increase.
- (ii) Since more current flows to the lamp, hence potential drop will increase and voltmeter reading will increase.

Q14. (a) An em wave is travelling in a medium with a velocity $\vec{v} = v\hat{i}$. Draw a sketch showing the propagation of the em wave, indicating the direction of the oscillating electric and magnetic fields.

(b) How are the magnitudes of the electric and magnetic fields related to the velocity of the em wave?

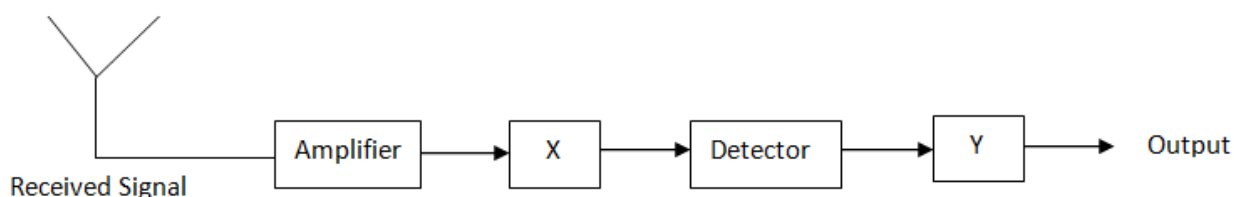
Answer:

	(a) We know electric and magnetic vectors vibrate perpendicular to each other and propagation of em wave is perpendicular to the plane containing electric and magnetic vector. Therefore since propagation is along X axis, as per right handed system of axis Electric vector along Y axis and Magnetic vector along Z axis as shown. (b) Velocity of em wave is ratio of magnitude of electric vector to the magnitude of magnetic vector. $c = \frac{ \vec{E}_m }{ \vec{B}_m }$
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Q15. Block diagram of a receiver is shown in the figure:

Receiving Antenna



- (a) Identify 'X' and 'Y'.
- (b) Write their functions.

Answer: (a)

X is intermediate frequency amplifier

Y is audio frequency amplifier

(b) Function of X (IF amplifier): The stages of an intermediate frequency amplifier are tuned to a fixed frequency that does not change as the receiving frequency changes. Typically, the IF centre frequency is chosen to be less than the desired reception frequency. The choice has some performance advantages e.g. it is easier and less expensive to get high selectivity at a lower frequency, secondly it is easier and less expensive to get high gain at a lower frequency. Usually the intermediate frequency is lower than the reception frequency, but in some modern receivers (e.g. scanners and spectrum analyzers) a higher IF frequency is used to minimize problems with image rejection or gain the benefits of fixed-tuned stages. The fixed frequency simplifies optimization of the IF amplifier. Intermediate-Frequency (IF) amplifiers are used to raise the signal level to a level that can be used by the decision making circuits to properly utilize the received information. While raising the signal level, they must also remove adjacent channel interference and images with filters so that only the desired signal passes. The constant frequency of the down converted signal is called the intermediate frequency (IF), and it is this signal that is processed by the intermediate-frequency amplifier.

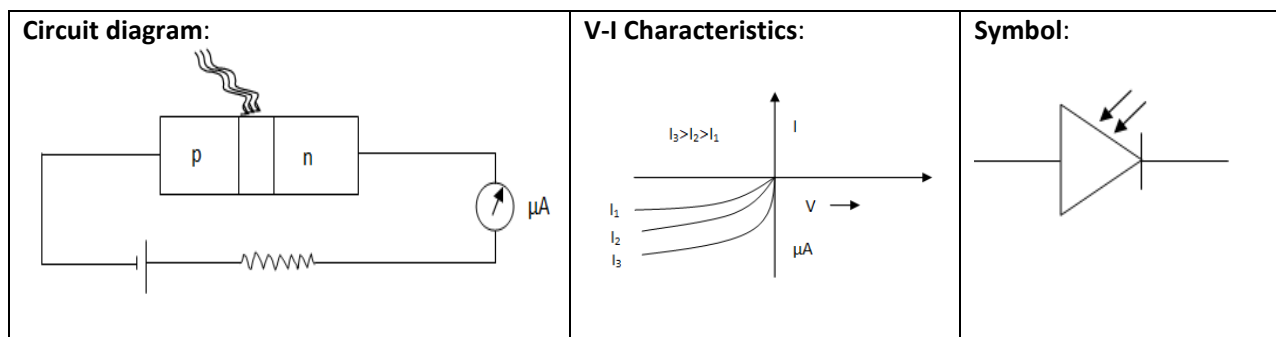
Audio frequency Amplifier: An amplifier is used to increase the amplitude of a signal waveform, without changing other parameters of the waveform such as frequency or wave shape.



Q16. Explain, with the help of a circuit diagram, the working of a photo-diode. Write briefly how it is used to detect the optical signals.

Answer:

Photodiode: It is a p-n junction with transparent window which allows light photons to incident on it. A photo diode is operated in reverse bias condition.



Working: When a photon of sufficient energy enters the depletion region of a semiconductor diode, it may strike an atom with sufficient energy to release the electron from the atomic structure. This creates a free electron and a hole (i.e. an atom with a space for an electron). The electron is negatively charged, while the hole is positively charged. The electrons and holes may remain free, or other electrons may combine with holes to form complete atoms again in the crystal lattice. However it is possible that the electrons and holes may remain free and be pulled away from the depletion region by an external field. In this way the current through the diode will change and a photocurrent is produced.

It is found that the current is proportional to the amount of light entering the intrinsic region. The more light, the greater the numbers of hole electron pairs that are generated and the greater the current flowing.

When not exposed to light the photo diode follows a normal V-I characteristic expected of a diode. In the reverse direction virtually no current flows, but in the forward direction it steadily increases, especially after the knee or turn on voltage is reached. This is modified in the presence of light. When used as a photo-diode it can be seen that the greatest effect is seen in the reverse direction. Here the largest changes are observed. This property is used in detecting optical or light signal.



OR

Mention the important considerations required while fabricating a p-n junction diode to be used as a Light Emitting Diode (LED). What should be the order of band gap of an LED if it is required to emit light in the visible range?

Answer:

Required consideration of p-n junction diode to be used as LED

- (1) It should emit maximum light.
- (2) It should not be operated a very high reverse voltage

Band gap required in LED to emit light in visible range is 1.8 eV.

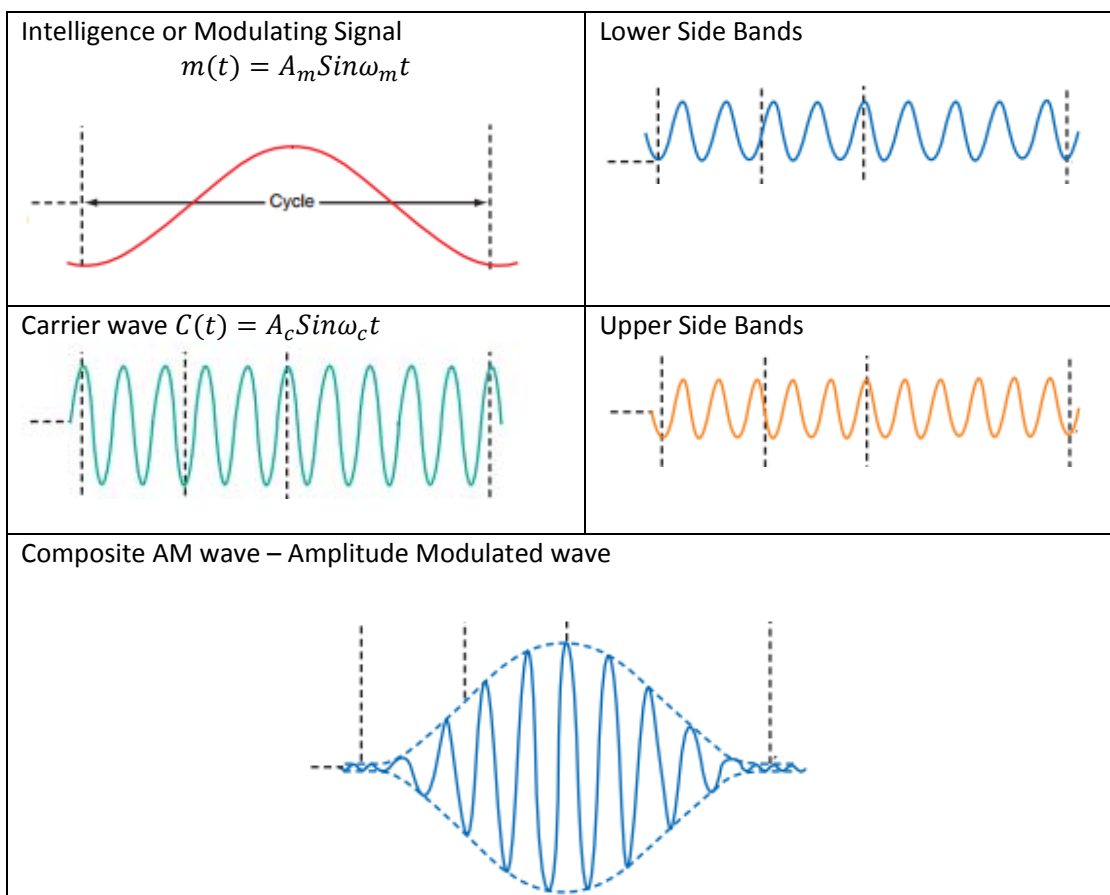
Q17. Write three important factors which justify the need of modulating a message signal. Show diagrammatically how an amplitude modulate wave is obtained when a modulating signal is superimposed on a carrier wave.

Answer: Three important reasons for modulating a message signal is

- i. For sending message signal to long distance without loss - We modulate the carrier signal with your information signal (signal which is to be transmitted) and then use a transmitter to send so that your information can travel to a longer distance. At the receiver side of the communication system the demodulator separates the information signal from the carrier and then your information signal is processed, in absence of modulation message signal will attenuate and cannot be send to long distance.
- ii. Antenna Size: In signal transmission antenna size required is one-fourth of wave length $h = \frac{\lambda}{4}$, thus to transmit 60 KHz signal ($\nu = 60 \times 1000 \text{ Hz}$, $\lambda = \frac{3 \times 10^8}{60 \times 1000} = 0.5 \times 10000 \text{ m} = 5 \text{ KM}$, $h = \frac{5}{4} = 1.25 \text{ KM}$), we require 1.25 KM antenna length, which is impractical to design and use. Modulation technique helps to reduce the frequency at the source hence low height antenna is required.
- iii. Modulation allows us to send a signal over a bandpass frequency range. If every signal gets its own frequency range, then we can transmit multiple signals *simultaneously* over a single channel, all using different frequency ranges.



Diagrammatic representation of Amplitude Modulation





Q18. A capacitor of unknown capacitance is connected across a battery of V volts. The charge stored in it is $360 \mu\text{C}$. When potential across the capacitor is reduced by 120 V , the charge stored in it becomes $120 \mu\text{C}$.

Calculate:

- (i) The potential V and the unknown capacitance C .
- (ii) What will be the charge stored in the capacitor, if the voltage applied had increased by 120V ?

Answer: Case I: $Q = 360 \mu\text{C}$, $Q=CV$, $360 = CV \rightarrow (1)$

Case II: $Q = 120 \mu\text{C}$, Potential = $V-120$, therefore $120 = C(V-120) \rightarrow (2)$

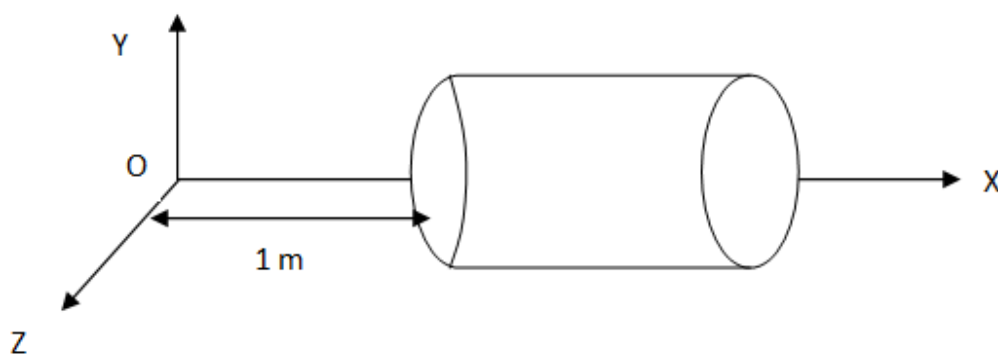
Or $120 = CV - 120C$, putting $CV=360$ from equation (1) we get $120 = 360 - 120C$ or $C = 240/120 = 2$

Also from 1, $360 = 2V$ or $V=180$.

- (i) Potential $V = 180 \text{ Volt}$, Capacitance $C = 2 \mu\text{f}$
- (ii) $V = 120$, $C = 2$, $Q = ?$ We know $Q=CV$, $Q = 2 \times 120 = 240 \mu\text{C}$

OR

A hollow cylindrical box of length 1m and area of cross-section 25 cm^2 is placed in a three dimensional coordinate system as shown in the figure.

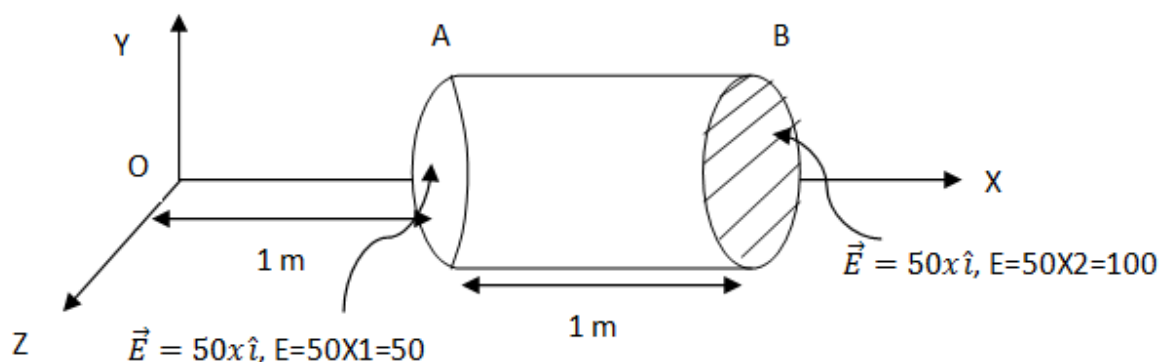


The electric field in the region is given by $\vec{E} = 50x\hat{i}$, where E is in NC^{-1} and x is in metres. Find:

- (i) Net flux through the cylinder.
- (ii) Charge enclosed by the cylinder.



Answer: Cylinder has one curved surface; electric lines are parallel to curved surface so flux is zero and two vertical surface A and B as shown in this figure



We know that flux = $\phi = \vec{E} \cdot \vec{A} = EA \cos \theta$

Total Flux = Flux through A + Flux through Curved Surface + Flux through B

$$= E_A A \cos 0 + E A \cos 90 + E_B A \cos 180$$

$$A=25, E_A=50, E_B=100$$

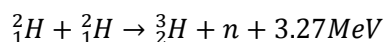
$$(i) \quad \text{Total Flux } \phi = 50 \times 25 + 0 + 100 \times 25(-1) = 1250 - 2500 = -1250$$

$$(ii) \quad \text{From Gauss's theorem } \phi = \frac{Q}{\epsilon_0} \text{ where } Q = \text{Enclosed charge}$$

$$\text{Therefore } Q = \epsilon_0 \phi = 8.85 \times 10^{-12} \times 1250 = 11062.5 \times 10^{-12} = 110.625 \times 10^{-10}$$



Q19. (a) In a typical nuclear reaction, e.g.



Although number of nucleons is conserved, yet energy is released. How? Explain.

(b) Show that nuclear density in a given nucleus is independent of mass number A.

Answer: (a) Right hand side has helium and neutron, Left hand side has two deuterons the sum of mass of right hand side component that is Helium and Neutron is less than the sum of two deuterons. This difference of mass called mass defect is emitted in the form of energy by Einstein mass energy equivalence formula $E = \Delta mc^2$, where $\Delta m = \text{mass defect}$, $c = \text{velocity of light}$.

(b) Nuclear density is the density of the nucleus of an atom. The nuclear density for a typical nucleus can be approximately calculated from the size of the nucleus, which itself can be approximated based on the number of protons and neutrons in it.

$$\text{Thus Nuclear Density}(d) = \frac{\text{Mass of Nucleus}}{\text{Volume}}$$

Given: m = Mass per nucleon

A = Mass number

R = Radius of Nucleus = $r_0 A^{1/3}$, $r_0 = 1.25\text{fm}$

$$d = \frac{mA}{\frac{4}{3}\pi R^3} = \frac{mA}{\frac{4}{3}\pi (r_0 A^{1/3})^3} = \frac{3mA}{4\pi r_0^3 A} = \frac{3m}{4\pi r_0^3}$$

Thus from the final expression ($\frac{3m}{4\pi r_0^3}$), we find nuclear density is independent of mass number (A).



Q20. (a) Why photoelectric effect cannot be explained on the basis of wave nature of light?
Give reasons.

(b) Write the basic features of photon picture of electromagnetic radiation on which Einstein's photoelectric equation is based.

Answer: (A) As per wave nature of light, intensity is the measure of energy carried by light. Therefore higher intensity light falling on metal surface should impart higher energy to electrons and ejected electrons should have higher Kinetic Energy, however in photo electric effect we know Kinetic energy of emitted electron is independent of Intensity of incident light.

(B) We know if the frequency of incident light is less than a minimum required frequency called threshold frequency, electron will not emit. This threshold frequency cannot be explained by wave nature of light.

(C) There is absolutely no time gap between incidence of light and emission of electron. However in classical wave theory we know that imparting energy to electrons and distribution of energy to atomic electrons should take some time, thus instantaneous energy emission in photo electric effect cannot be explained by wave nature of light.

(b) **Basic feature of photon**

(i) Light energy carrier is not ordinary particle but quantum mechanical particle called photon, ordinary particle has non zero rest mass (i.e. mass of particle when velocity is zero), rest mass of photon is zero

$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$, m = mass of photon when moving with velocity v , m_0 = rest mass of photon.

(ii) Energy of each photon is given by $E = h\nu$, where frequency $\nu = \frac{c}{\lambda}$, h = Planks constant

(iii) Incident photon energy is used to eject electron (work function) and remaining part converted as Kinetic energy of ejected electron. $E = h\nu + \frac{1}{2}mv^2$

(iv) Electron emission depends on frequency of incident light (photon) not on Intensity of light.

(v) There is a minimum frequency of photon which if satisfied called threshold frequency then only electrons will come out from metal surface on collision with photon.



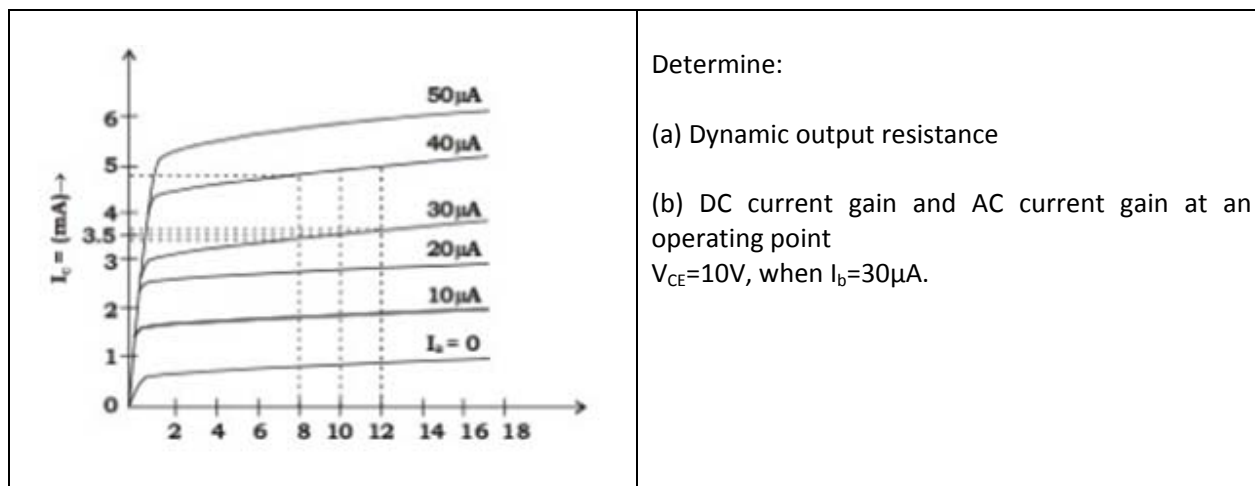
Q21. A metallic rod of length 'l' is rotated with a frequency ν with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius r , about an axis passing through the centre and perpendicular to the plane of the ring. A constant uniform magnetic field B parallel to the axis is present everywhere. Using Lorentz force, explain how emf is induced between the centre and the metallic ring and hence obtain the expression for it.

Answer: Free electrons of the rotating rod moves outward due to Lorentz force and distributed on the circumference of the ring, this separation creates emf across the terminal of the rod.

	Given B = Magnetic Field, ν = frequency r = radius Let dE be the emf produced across length dr then $dE = BVdr = Br\omega dr$ therefore total emf can be obtained by integrating dE $\int dE = \int_0^r B\omega r dr = B\omega \frac{r^2}{2} = B2\pi\nu \frac{r^2}{2} = \pi\nu Br^2$
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Q22. Output characteristics of an n-p-n transistor in CE configuration is shown in the figure.



Answer:

(a) Dynamic output resistance $r_o = \left(\frac{\Delta V_{CE}}{\Delta I_C} \right)_{I_b} = \left[\frac{12-8}{(3.6-3.4) \times 10^{-3}} \right]_{30\mu A} = \left[\frac{4000}{0.2} \right]_{30\mu A} = 20000\Omega = 20K\Omega$

(b) DC Current Gain $\beta_{dc} = \frac{I_C}{I_b} = \frac{3.5 \times 10^{-3}}{30 \times 10^{-6}} = 0.116 \times 10^3 = 116$

AC Current Gain $\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_b} \right)_{V_{CE}}$

From the figure above when base current is $30\mu A$, collector current is 3.5 mA also when base current is $40\mu A$, collector current is 4.6 mA .

Therefore collect current change $\Delta I_C = 4.6 - 3.5 = 1.1\text{ mA} = 1.1 \times 10^{-3}\text{ A}$

Base current change $\Delta I_b = 40 - 30 = 10\mu A = 10 \times 10^{-6}\text{ A}$

Therefore $\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_b} \right)_{V_{CE}} = \frac{1.1 \times 10^{-3}}{10 \times 10^{-6}} = 110$



Q23. Using Bohr's postulates, obtain the expression for the total energy of the electron in the stationary states of the hydrogen atom. Hence draw the energy level diagram showing how the line spectra corresponding to Balmer series occur due to transition between energy levels.

Answer: Total Energy of electron in hydrogen atom = K.E + P.E

Centripetal force is due to the electrostatic attractive force between nucleus and electron

$$F = \frac{mv^2}{r} \rightarrow (1)$$

Where m= mass of electron

v= velocity of electron

r= radius of the orbit in which electron rotates.

$$\text{Electrostatic force } F = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{r^2} \rightarrow (2)$$

From equation (1) and (2)

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{r^2} \rightarrow (3)$$

$$\text{Or } K.E = \frac{mv^2}{2} = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{2r} \rightarrow (4)$$

As per Bohr's postulates electron has discrete energy state: Electron rotates in non radiating orbits where its angular momentum is integral multiple of $\frac{h}{2\pi}$

$$mvr = n \frac{h}{2\pi}$$

$$v = \frac{nh}{2\pi mr} \rightarrow (5)$$

From equation (3) and (5)

$$\frac{m \left(\frac{nh}{2\pi mr} \right)^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{r^2}$$

$$\frac{mn^2h^2}{r^4\pi^2m^2r^2} = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{r^2}$$

$$\frac{n^2h^2}{r\pi m} = \frac{Ze.e}{\epsilon_0} \text{ or } r = \frac{n^2h^2\epsilon_0}{\pi mZe^2} \text{ for H atom } Z = 1, \text{ therefore } r = \frac{n^2h^2\epsilon_0}{\pi me^2} \rightarrow (6)$$



Electric Potential Energy= Potential X Charge

$$P.E = \frac{1}{4\pi\epsilon_0} \frac{Ze}{r} \times (-e) = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} \rightarrow (7)$$

Therefore Total Energy = K.E + P.E putting values from equation (4) and (7) we get

$$\text{Total Energy} = \frac{1}{4\pi\epsilon_0} \frac{Ze.e}{2r} - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{2r}$$

Putting the value of r from equation (6)

$$\text{Total Energy} = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{\frac{n^2 h^2 \epsilon_0}{\pi m e^2}}, \text{ putting } Z=1$$

$$\text{Total Energy} = -\frac{m e^4}{4 \epsilon_0^2 n^2 h^2} \text{ Putting the values of } m, e, h \text{ and } \epsilon_0 \text{ we get Total Energy} = -\frac{13.6}{n^2}$$

Energy Level Diagram for Spectral Line of Balmer Series: Electron transition for higher level ($n > 2$) and terminating at $n=2$ gives Balmer Series spectral line.

	Therefore the energy difference when electron transit to 2nd orbit ($n=2$) $\Delta E = E_2 - E_n = -13.6 \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$ Where $n > 2$, $n=3, 4, 5 \dots$ In 2nd orbit energy of electron $E_2 = -\frac{13.6}{n^2} = -\frac{13.6}{2^2} = -3.4 \text{ eV}$
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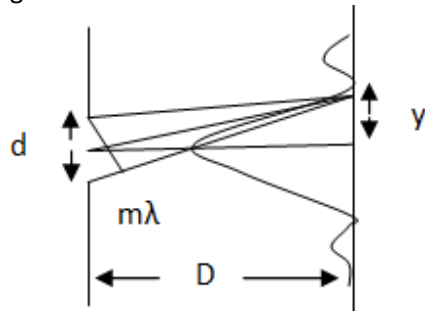


Q24. (a) In what way is diffraction from each slit related to the interference pattern in a double slit experiment?

(b) Two wavelength of sodium light 590 nm and 596 nm are used, in turn, to study the diffraction taking place at a single slit of aperture 2×10^{-4} m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the positions of the first maxima of the diffraction pattern obtained in the two cases.

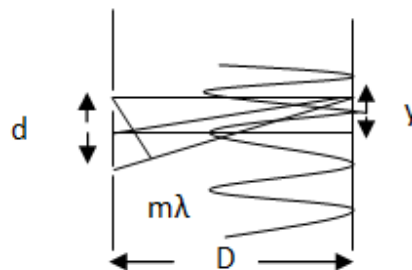
Answer: (a)

Single slit diffraction pattern: A path difference of 1 wavelength produces a minimum since the light from the centre of the slit differs by half a wavelength this gives destructive interference

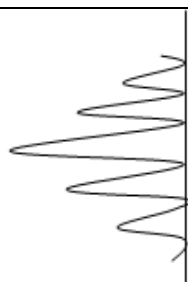


Condition for minimum $y = \frac{m\lambda D}{d}$

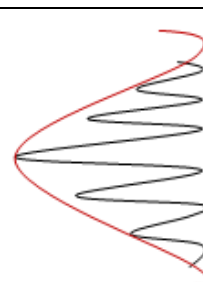
In double slit interference : path difference of 1 wavelength produces constructive interference



Condition for maximum $y = \frac{m\lambda D}{d}$



Actual double slit interference pattern



Envelop shows single slit diffraction



(b) Given $d = 2 \times 10^{-4} \text{ m}$, $D = 1.5 \text{ m}$, $\lambda_1 = 590 \times 10^{-9} \text{ m}$, $\lambda_2 = 596 \times 10^{-9} \text{ m}$

For first maxima

$$d \sin \theta = \frac{3\lambda_1}{2}, \text{ since } \theta \text{ is very small } \sin \theta \approx \theta \approx \tan \theta$$

$$\text{or } d \tan \theta = \frac{3\lambda_1}{2} \text{ or } d \cdot \frac{y_1}{D} = \frac{3\lambda_1}{2} \text{ or } y_1 = \frac{3\lambda_1 D}{2d} \text{ for the other wavelength } y_2 = \frac{3\lambda_2 D}{2d}$$

$$\text{Therefore the path difference } \Delta y = \frac{3D}{2d} (\lambda_2 - \lambda_1) = \frac{3 \times 1.5 \times (596 - 590) \times 10^{-9}}{2 \times 2 \times 10^{-4}} = 6.75 \times 10^{-4} \text{ m}$$

Q25. In a series LCR circuit connected to an ac source of variable frequency and voltage $v = v_m \sin \omega t$, draw a plot showing the variation of current (I) with angular frequency (ω) for two different values of resistance R_1 and R_2 ($R_1 > R_2$). Write the condition under which the phenomenon of resonance occurs. For which values of the resistance out of the two curves, a sharper resonance is produced? Define Q-factor of the circuit and give its significance.

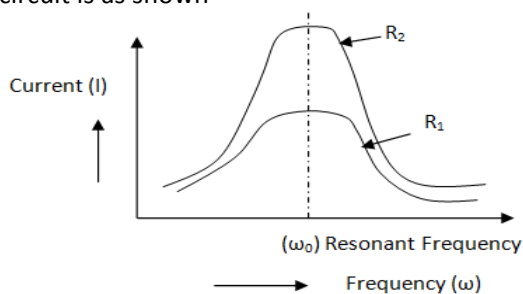
Answer:

(i) We know that rms current in LCR circuit is given by

$$I = \frac{E}{\sqrt{R^2 - \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

If $R_1 > R_2$ then current for R_1 will be less i.e. $I_1 < I_2$.

Variation of current with angular frequency in LCR circuit is as shown



(ii) Condition for resonance Also for current to be maximum, denominator should be minimum possible when

$$\omega L = \frac{1}{\omega C}$$

That is resistance offered to the ac by inductance should be equal to the resistance offered by capacitor. Also resonant frequency

$$\omega = \frac{1}{\sqrt{LC}} = \omega_0$$

(iii) Sharper resonance is produced by curve of R_2 , the curve peak is not flattened to much as compared to R_1 .

(iv) Q-factor: Sharpness of resonance is decided by Q-factor and is defined as ratio of resonant frequency to the difference of half power frequency, mathematically

$$Q - \text{factor} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Physical Significance: since $R_1 > R_2$ therefore Q-factor of R_1 is less than Q-factor of R_2 hence sharpness of R_2 is more.



Q26. While travelling back to his residence in the car, Dr. Pathak was caught up in a thunderstorm. It became very dark. He stopped driving the car and waited for thunderstorm to stop. Suddenly he noticed a child walking alone on the road. He asked the boy at his residence. The boy insisted that Dr. Pathak should meet his parents. The parents expressed their gratitude to Dr. Pathak for his concern for safety of the child.

Answer the following questions based on the above information:

- (a) Why is it safer to sit inside a car during a thunderstorm?
- (b) Which two values are displayed by Dr. Pathak in his actions?
- (c) Which values are reflected in parents' response to Dr. Pathak?
- (d) Give an example of a similar action on your part in the past from everyday life.

Answer: (a) We know that electric field does not exist inside a conductor (sphere), here since outside there is a thunderstorm it is safer to stay inside a car so that the car provides a shield to the electrostatic field.

(b) Dr. Pathak has good knowledge of physics (i) he is very social, gives value to human life (ii) helping attitude to have concern for other safety.

(c) Gratitude, giving thanks to the person who helped.

(d) There was a football match going on, suddenly heavy rain with a thunderstorm started, to protect us from rain initially we took shelter under a tree, one of our team members realised it is not safe to stay under a tree due to a thunderstorm as electricity can easily reach the ground through a long tree, all of us rushed to the parking area and took shelter inside a bus which brought us home.

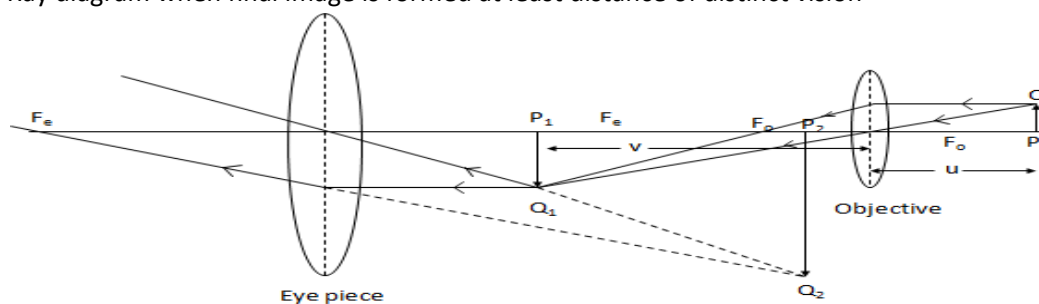


Q27. (a) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain expression for total magnification when the image is formed at infinity.

(b) Distinguish between myopia and hypermetropia. Show diagrammatically how these defects can be corrected.

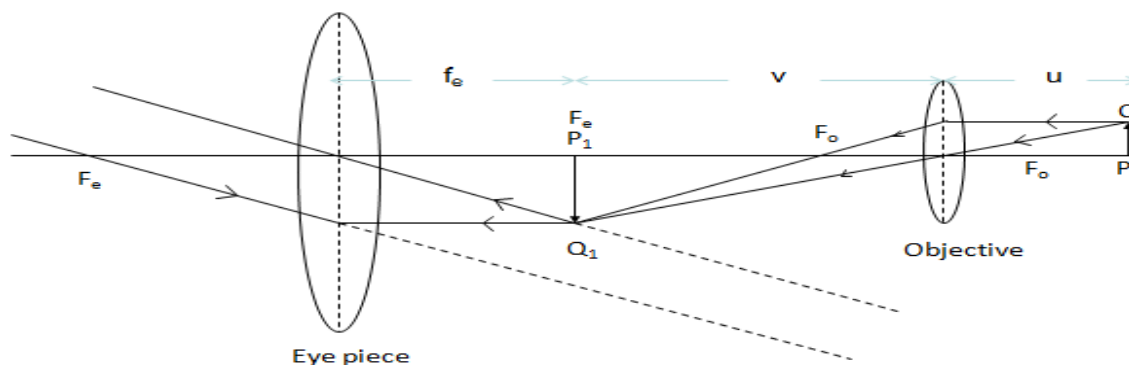
Answer: (a) Compound Microscope: It consists of objective and eye piece. The objective is a convex lens of small aperture and small focal length. The aperture of the eye piece lens is slightly greater and the focal length is also small.

Ray diagram when final image is formed at least distance of distinct vision



PQ is an object kept in front of objective ($f_o < u < 2f_o$), P_1Q_1 is the image formed by objective this image serves as object for eye piece, since P_1Q_1 lies between the focus and optical centre of eye piece hence an extended image P_2Q_2 is formed by eye piece. Thus P_2Q_2 is the final image of PQ produced by the compound microscope.

Ray diagram when final image is formed at infinity: Here the position of eye piece is so adjusted that image formed by objective i.e. P_1Q_1 falls at the focal plane of eye piece, hence final image is formed at infinity





Magnifying Power (M) when image is formed at infinity:

$$M = \text{Magnifying Power of Objective } (M_o) \times \text{Magnifying Power of Eyepiece } (M_e)$$

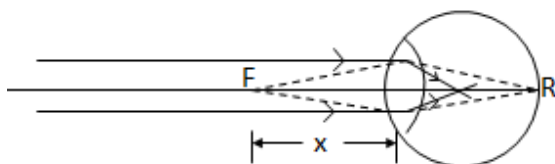
$$M_o = \frac{v}{u} = \frac{\text{Image distance}}{\text{Object distance}}, M_e = \frac{\text{Image distance}}{\text{Object distance}} = \frac{D}{f_e},$$

Image is actually formed at infinity but seen at least distance of distinct vision (D)

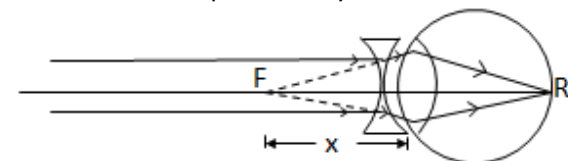
$$M = \frac{v}{u} \times \frac{D}{f_e}$$

(b)

Myopia: Short sighted people cannot see distance objects distinctly, due to some reason the normal (maximum) focal length of eye lens decreases i.e. the power of eye lens increases, hence parallel rays from distance object come to focus in front of the retina and this is why it can't be seen. As the object comes closer to the eye from infinity image shifts back and when object comes to F, the image falls at the retina and can be seen. Hence F is the far point for the defective eye.



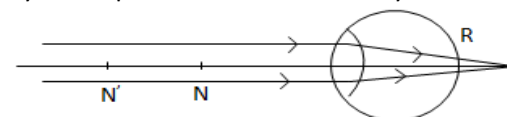
Remedy: Since the normal power of eye lens has increased hence this defect can be corrected for, by using an auxiliary lens in front of the eye lens which will decrease the power of eye lens.



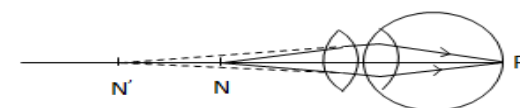
The auxiliary lens will create the image of the distant object at F and this image serves as object for eye lens produce image at the retina and thus the distant object can be seen.

Hypermetropia: Long sighted people can see distant objects distinctly but cannot see nearer objects. Due to some reason normal focal length of eye lens increases i.e. normal power p_1 of eye lens decreases and hence parallel rays from distant object come To focus behind the retina. Hence to see distant objects focal length of eye lens is decreased by exerting pressure with the ciliary muscles i.e. accommodation starts right from infinity.

As the objects moves towards the eye, accommodation continues since it has been started earlier it will be exhausted earlier that is it will get exhausted at N' , much before reaching the near point N for the normal eye. Thus for this defective eye near point has shifted from eye.



Remedy: Since the normal power P_1 of the eye lens has decreased hence to increase the equivalent power P ($P > P_1$) a power $+P_2$ should be added to P_1 i.e. and auxiliary lens convex in nature should be used.



The auxiliary lens produces the image of the object kept at N (near point for the normal eye) at N' this image serves as object for the eye lens, image is now produced at retina by eye lens. Therefore object kept at N can be seen.



Myopia correction: Calculation: Let x = distance of the far point for the defective eye. f = the focal length of the auxiliary lens For the auxiliary lens : $u = \infty$, $v = -x$ Using the formula for the lens : $\frac{1}{\infty} + \frac{1}{-x} = \frac{1}{f}$ $\therefore f = -x$ The negative sign indicates that the nature of the auxiliary lens is concave.	Hypermetropia correction: Calculation: Given x & D = least distance of distinct vision for the defective eye and the normal eye respectively. Let f = the focal length of the auxiliary lens. $\frac{1}{-x} + \frac{1}{D} = \frac{1}{f}$ $\frac{1}{D} - \frac{1}{x} = \frac{1}{f}$ $f = \frac{x D}{x - D}$ $\therefore x > D$ f is positive, the auxiliary lens is convex.
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OR

(a) State Huygen's principle. Using this principle draw a diagram to show how a plane wave front incident at the interface of the two media gets refracted when it propagates from a rarer to a denser medium. Hence verify Snell's law of refraction.

(b) When monochromatic light travels from a rarer to a denser medium, explain the following, giving reasons:

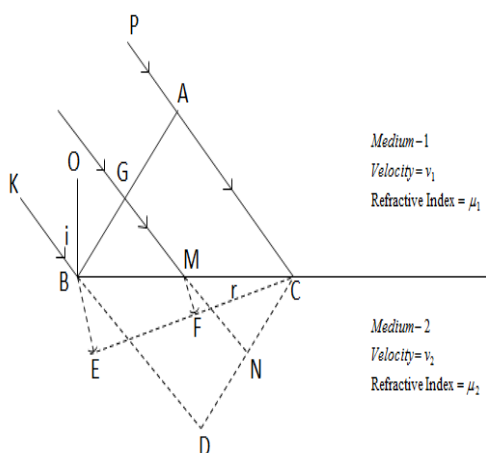
(i) Is the frequency of reflected and reflected light same as the frequency of incident light?

(ii) Does the decrease in speed imply a reduction in the energy carried by light wave?

Answer : (a) Huygen's Principle:

(i) Every point on a given wave front acts as a fresh source of secondary wavelets which travel in all directions with the speed of light.

(ii) The forward envelope of these secondary wavelets gives the new wave front at any instant.



BC is the section of the refracting surface by the plane of the paper i.e. the out face, separating two media of refractive indices μ_1 and μ_2 ($\mu_1 < \mu_2$)

AB is the section of the incident plane wave front by the plane of the paper at an instant of time $t=0$

Both the refracting surface and the plane wave front are perpendicular to the plane of the paper.

v_1 & v_2 = Velocity of light in medium-1 and medium-2 respectively.

PA and KB are perpendiculars dropped on the incident wave front AB and hence they are the incident rays. BO is normal drawn at B on the refracting surface.

$$\angle KBO = i = \text{angle of incidence}$$

$$\angle KBA = 90^\circ \therefore \angle OBA = 90^\circ - i$$

$$\angle OBC = 90^\circ \therefore \angle ABC = 90^\circ - (90^\circ - i) = i$$



Hence angle of incidence can also be defined as the angle between the incident wave front and the refracting surface. Similarly the angle of refraction can also be defined as the angle between the refracted wave front and the refracting surface.

Had the two medium been same, then the rays from the point G & B would have gone up to N & D respectively during the time t in which ray from a goes up to C so that

$$AC=GN=BD=v_1t \quad (1)$$

Hence CND would have been the position of the wave front after time t . But since the two medium are different let the ray from B go up to E in time t . So that

$$t = \frac{AC}{v_1} = \frac{BE}{v_2} \rightarrow (2)$$

With B as center and BE as radius we imagine a sphere the section of which by the plane of the paper gives an arc of a circle. CE is drawn tangential to that arc. Then a plane through CE and perpendicular to the plane of the paper would represent the refracted wave front after time t provided we can show that ray from any other point G on the incident wave front also touches the plane through CE after time t . To prove it, from M, MF is drawn perpendicular to CE.

$$\triangle BDC \text{ \& \; } \triangle MNC \text{ are similar } \therefore \frac{BD}{MN} = \frac{BC}{MC} \rightarrow (3)$$

$$\triangle BEC \text{ \& \; } \triangle MFC \text{ are similar } \therefore \frac{BE}{MF} = \frac{BC}{MC} \rightarrow (4)$$

$$\text{From (3) \& (4)} \quad \frac{BD}{MN} = \frac{BE}{MF}$$

$$\text{Putting equation (1) \& (2)} \quad \frac{v_1 t}{MN} = \frac{v_2 t}{MF} \text{ or } \frac{MF}{v_2} = \frac{MN}{v_1}$$

That is time taken to go from M to F in medium 2 and time taken to go from M to N in medium 1 is same.

Hence the time taken by the light to go from G to M + M to N is same as the time taken to go from G to M + M to F i.e. from G to F.

Hence CE is the refracted wave front after time t .



$\therefore \angle BCE$ = angle between the refracting surface and the
refracted wave front = angle of refraction = r (using definition)

We now prove Snell's law :

$$\text{From } \triangle ABC : \sin i = \frac{AC}{BC}$$

$$\text{From } \triangle ABC : \sin r = \frac{BE}{BC}$$

$$\therefore \frac{\sin i}{\sin r} = \frac{AC/BC}{BE/BC} = \frac{AC}{BE}$$

$$\text{Putting equation (1) and (2): } \frac{\sin i}{\sin r} = \frac{v_1 t}{v_2 t}$$

$$\frac{\sin i}{v_1} = \frac{\sin r}{v_2}$$

Multiplying both sides by v_0 where v_0 is the velocity of light in vacuum.

$$\frac{v_0}{v_1} \sin i = \frac{v_0}{v_2} \sin r$$

$$\mu_1 \sin i = \mu_2 \sin r \quad \text{First law proved}$$

Thus above result verifies Snell's law.

From the nature of construction it is evident that the incident ray refracted ray & normal at the point of incidence all lay on the same plane i.e. on the plane of the paper.

(b) (i) Frequency of light never changes when light passes from one medium to other medium. It is the fundamental property of light.

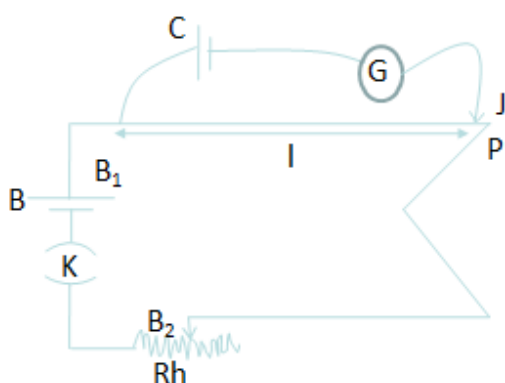
(ii) Energy of light given by $E = h\nu$, since ν i.e. frequency remains unchanged hence energy of light never changes while passing from one medium to other and it is independent of speed.



Q28. (a) State the working principle of a potentiometer. With the help of the circuit diagram; explain how a potentiometer is used to compare the emf's of two primary cells. Obtain the required expression used for comparing the emf's.

(b) Write two possible causes for one sided deflection in a potentiometer experiment.

Answer: (a) Working Principle of Potentiometer: The potential drop across the potentiometer wire is directly proportional to the length provided the cross-sectional wire is uniform and constant current flows through potentiometer wire.



If $E = V_{B_1P}$ no current flows through the galvanometer and galvanometer shows null deflection.

Let i be the current flowing through the potentiometer wire by the driving cell.

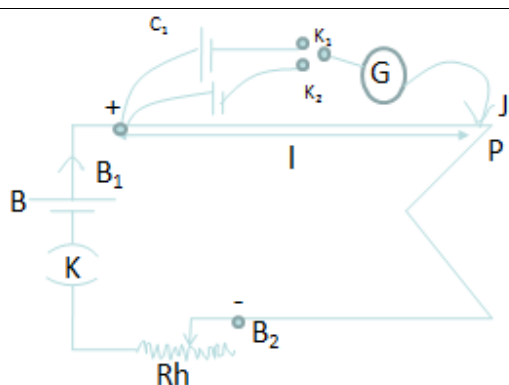
ρ = The resistance per unit length throughout the potentiometer wire.

$l = B_1P$ = The length of the potentiometer wire up to the jockey at null deflection.

Therefore the potential difference across the wire B_1P

$$V_{B_1P} = i\rho l$$

E = emf of the test cell.



Comparing emf of two cells: The positive terminal of both the cells for which emf to be compared is connected to B_1 and negative of the test cells are connected through a key and galvanometer as shown.

If K_1 is closed (K_2 is open) balance point is found by using C_1

$$E_1 = i\rho l_1$$

If K_1 is opened and K_2 is closed by measuring C_2 balance point is found

$$E_2 = i\rho l_2$$

$$\frac{E_1}{E_2} = \frac{i\rho l_1}{i\rho l_2} = \frac{l_1}{l_2}$$

Thus the ratio of balance length is the ratio of emf of cells.



(b) Reason for one sided deflection across the potentiometer wire

- (i) The potential drop across the whole length of the potentiometer wire is less than the emf of the test cell.
- (ii) Test cells positive terminal is not connected to the potentiometer circuit where the driving cell positive terminal is connected.

OR

(a) State Kirchhoff's rules for an electric network. Using Kirchhoff's rules, obtain the balance condition in terms of the resistances of four arms of Wheatstone bridge.

	(b) In the meter bridge experimental set up, shown in the figure, the null point 'D' is obtained at a distance of 40 cm from end A of the meter bridge wire. If a resistance of $10\ \Omega$ is connected in series with R_1, null point is obtained at AD=60 cm. Calculate the values of R_1 and R_2.
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Answer: In complicated networks to find current flowing through different branches we apply Kirchhoff's law.

First law: The algebraic sum of currents meeting at a point is zero. i.e. $\sum i = 0$.

In the given circuit, Considering outward direction of current as positive and inward as negative then at the point F

According to Kirchhoff's first law, at junction F

$$I_1 + I_2 = I_3$$

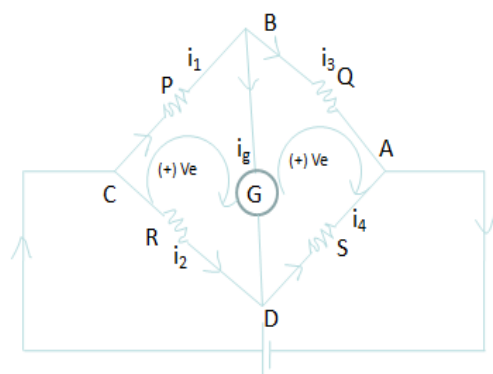
Second law: In a closed mesh of electrical conductors the algebraic sum of product of resistance and the respective current in the different branches is equal to the total e.m.f applied in the closed mesh.

$$\sum iR = \sum E$$



Using Kirchhoff's law we now find the condition for null deflection in a Wheatstone bridge

Four resistance P, Q, R and S are joined end to end to form a closed circuit. These close networks of conductors form a closed circuit. This close network of conductors is known as Wheatstone bridge. Between any pair of opposite junctions say A & C a battery is connected and between the other pair of opposite junctions a galvanometer is connected. We assigned the current flowing in the different branches from the logical consideration.



Given : G = The resistance of the galvanometer
Let i_1, i_2, i_3, i_4 & i_g be the current flowing through the branches of resistance P, R, Q, S and G respectively.

Applying Kirchhoff's first law :

(1) At the point C :

$$\sum i = i - i_1 - i_2 = 0 \text{ or } i_2 = i - i_1 \rightarrow (1)$$

(2) At the point B :

$$\sum i = i_1 - i_3 - i_g = 0 \text{ or } i_3 = i_1 - i_g \rightarrow (2)$$

(3) At the point D :

$$\sum i = i_2 + i_g - i_4 = 0 \text{ or } i_4 = i - i_1 + i_g \rightarrow (3)$$

Applying Kirchhoff's second law :

In closed mesh CBDC :

$$i_1 P + i_g G - i_2 R = 0$$

$$\text{Putting equation(1) : } i_1 P + i_g G - (i - i_1) R = 0$$

$$\text{or } (P + R)i_1 + i_g G - iR = 0 \rightarrow (4)$$

(ii) In closed mesh BADB

$$i_3 Q - i_g G - i_4 S = 0 \rightarrow (5)$$

Putting equation(2) in equation (5)

$$(i_1 - i_g)Q - i_g G - (i - i_1 + i_g)S = 0$$

$$\text{or } i_1(Q + S) - i_g(Q + G + S) - iS = 0 \rightarrow (6)$$

To eliminate i_1 from the equations multiplying equation(4) by $(Q + S)$ and (6) by $(P + R)$ and subtracting

$$i_1(Q + S)(P + R) + i_g G(Q + S) - iR(Q + S) = 0$$

$$i_1(Q + S)(P + R) - i_g(P + R)(Q + G + S) - iS(P + R) = 0$$

$$i_g[G(Q + S) + (Q + S + G)(P + R)] - i[R(Q + S) - S(P + R)] = 0$$

$$i_g = \frac{i(RQ - SP)}{G(Q + S) + (Q + S + G)(P + R)} \rightarrow (6)$$

From equation (6) we can find the current flowing through the galvanometer. For null deflection $i_g = 0$ from equation (6) $i_g = 0$ only if $(RQ - PS) = 0$ or $P/Q = R/S$ is the condition for null deflection.



(b) $\frac{R_1}{R_2} = \frac{AD}{DC} = \frac{40}{60} \rightarrow (1)$ also $\frac{R_1+10}{R_2} = \frac{60}{40} \rightarrow (2)$

Dividing (2) by (1)

$$\frac{R_1+10}{R_1} = \frac{60}{40} \times \frac{60}{40} = \frac{9}{4}$$

$R_1 = 8 \Omega$

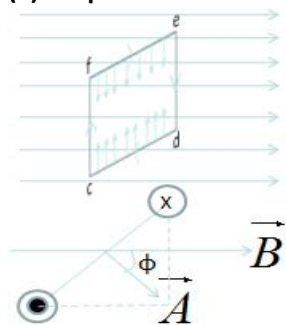
From (1): $R_2 = 12 \Omega$

Q29. (a) Derive the expression for the torque on a rectangular current carrying loop suspended in a uniform magnetic field.

(b) A proton and a deuteron having equal momenta enter in a region of uniform magnetic field at right angle to the direction of the field. Depict their trajectories in the field.

Answer:

(a) Torque on a current carrying coil



Theory : Given n = the number of turns in the coil

l & b = the length of each vertical & horizontal side of the coil respectively (assuming) the coil to be rectangular

i = the current flowing through the coil.

B = the induction vector of the magnetic field in which the coil is suspended.

We know that the force experienced by the current carrying conductor placed in a magnetic field is

$$\vec{F} = i(\vec{l} \times \vec{B}) \rightarrow (1)$$

Force on each wire 'de' and 'fc'

$$|\vec{F}| = ibB \sin(90 - \phi) = ibB \sin \phi$$

where ϕ = angle between the normal to the plane of the coil and the direction of the magnetic field.

The direction of the force on the wires 'de' & 'fc' are in the plane of the coil along the downward and upward direction respectively and hence cancel out.

The force on the two vertical wires 'cd' and 'ef' by using equation(1) are found to have magnitude

$$F = iB \sin 90^\circ = ilb$$

Applying the right hand curl rule for vector product the directions of the force are as shown in the figure. These two forces being equal in magnitude, opposite in direction parallel and non co-planer constitute a couple.



Moment of the couple $\Gamma = i\vec{B} \times \vec{OC}$

$$\Gamma = i\vec{B}b \sin \phi = i(lb)\vec{B} \sin \phi = i\vec{A}\vec{B} \sin \phi$$

Since there are n turns in the coil the total torque experienced by the current carrying coil in the magnetic field

$$\Gamma = n\vec{A}\vec{B} \sin \phi$$

(b)

When charge particle enters into the magnetic field, the force exerted by magnetic field on the charge provides the required centripetal force

$$\frac{mv^2}{r} = q(\vec{v} \times \vec{B}) \text{ or } r = \frac{mv}{qB}$$

Both deuteron and proton has same charge and velocity hence they will execute circular path of same radius.

OR

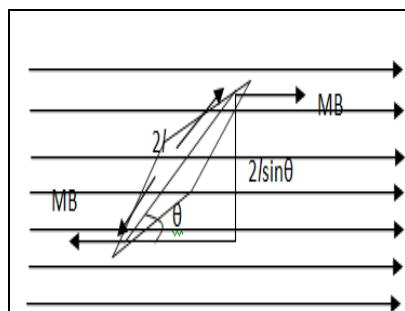
(a) A small compass needle of magnetic moment 'm' is free to turn about an axis perpendicular to the direction of uniform magnetic field 'B'. The moment of inertia of the needle about the axis is 'I'. The needle is slightly disturbed from its stable position and then released. Prove that it executes simple harmonic motion. Hence deduce the expression for its time period.

(b) A compass needle, free to turn in a vertical plane orients itself with its axis vertical at a certain place on the earth. Find out the values of (i) horizontal component of earth's magnetic field and (ii) angle of dip at the place.

Answer: (a) Given m =Magnetic moment= $M \times 2l$, where M =Pole strength, $2l$ =length

B = Uniform Magnetic field

I = Moment of inertia of the needle about the axis.



The magnetic field exerts force MB on north and south pole of the needle. If θ be the inclination of the needle with the magnetic field.

Then torque due to this equal and opposite force

$$\tau = MB \times 2l \sin \theta = mB \sin \theta \rightarrow (1) \text{ since } M \times 2l = m$$

$$\text{Therefore } I \times \frac{d^2 \theta}{dt^2} = -mB \sin \theta = -mB \theta \rightarrow (2) \sin \theta \approx \theta$$

$$\tau \propto -\frac{d^2 \theta}{dt^2}$$

Negative sign indicates restoring nature. Therefore the motion is SHM.

$$\text{Time period } T = 2\pi \sqrt{\frac{\text{angular displacement}}{\text{angular acceleration}}} = 2\pi \sqrt{\frac{I}{MB}} \rightarrow (3)$$

Equation (3) gives the time period of oscillation.



(b) (i) Since the needle is aligned vertically the horizontal component of earth's magnetic field is ZERO.

(ii) Resolving earth's magnetic field into horizontal and vertical component as shown.

$\vec{B}_H = \vec{B}_t \cos \delta$ $\vec{B}_v = \vec{B}_t \sin \delta$	$\vec{B}_t$ = Earth's magnetic field. $\vec{B}_H = \vec{B}_t \cos \delta$ = Horizontal component $\vec{B}_v = \vec{B}_t \sin \delta$ = Vertical component $\tan \delta = \frac{B_v}{B_H}$
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